

Motor Imagery Analysis Based on Filter Bank Common Spatial Pattern

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Abstract—This paper analyzes the EEG signals of left and right hand motor imagery and uses Filter Bank Common Spatial Pattern to extract features from EEG signals. Compared with Discrete Wavelet Transform, Autoregressive mode, Power Spectral Density and Common Spatial Pattern, it is found that FBCSP can significantly improve the recognition rate. While extracting the features of sub-frequency bands, this paper also considers ERD and ERS in motor imagery and the distribution of signal energy in different frequency bands. It is found that during the time of the motor imagery, the energy changes are generally concentrated on the μ rhythm, and the motor sensory area of the cerebral cortex has the nature of contralateral reflection of left and right hand motor imagery. Researchers generally use Support Vector Machine classifiers for pattern recognition classification, but the classification effect of a traditional SVM classifier is not ideal. Therefore, the paper uses the integration algorithm AdaBoost and algorithm Gradient Boosting for classification. Compared with Adaboost, the error rate of Gradient Boosting algorithm drops more sharply with the increased number of iterations, and the accuracy of Gradient Boosting algorithm is higher.

Keywords Brain-Computer Interface; motor imagery; FBCSP; Boosting

I. INTRODUCTION

Nowadays, Brain-machine interface (BCI) is of great concern for researchers. Under different perceptual awareness, different rhythms in EEG signals show different states of activity. Different actions or thoughts influence these rhythms. In motor imagery, feature extraction of EEG signals is an important part of motor imagery analysis. There are many common methods of feature extraction.

Wavelet transform is a signal time-scale analysis method [1]. It has the characteristics of multi-resolution analysis, and it has the ability to characterize the local characteristics of the signal in both time and frequency domains. The time-frequency localization does the analyses with a fixed window, and the time window and frequency window size can be changed [2]. The steps of Discrete Wavelet Transform include discretizing

the scale parameter according to the power series, and performing uniform discretization of time.

Autoregressive Model is the process of using the data itself as a regression variable [3]. To be specific, It uses the linear combination of random variables at several moments in the previous period to describe the linear regression model of random variables at a certain moment in the future. AR model can be used to identify the state of EEG signals [4].

Power Spectral Density expresses the distribution of signal power at each frequency point [5]. PSD uses the mean square value of random variables and the dimension of average power per unit frequency as the extracted features [6].

In order to better feature extraction, Common Spatial Pattern can select a set of optimal spatial filters for projection [7], so as to maximize the variance of the two kinds of signals and obtain feature vectors with high discrimination [8]. CSP is also called Fukunaga-Koontz transform in machine learning. It was firstly proposed by Fukunaga and Koontz. This is an improved algorithm of principal component analysis [9]. Since then, it has been widely used in various fields, such as digital and face recognition, target recognition, and abnormal detection of EEG signals. Later, CSP was successfully applied to brain-computer interfaces as a spatial filtering law [10], and was considered as effective method for extracting event-related desynchronization.

Considering about using CSP for feature extraction, the frequency bands are relatively wide. In order to improve the accuracy of feature extraction, this paper uses FBCSP to extract feature vectors. FBCSP performs frequency-domain filtering on preprocessed EEG data to obtain multiple sub-frequency bands, and carries out spatial filtering on each sub-frequency band using CSP method, and then selects features and classifies EEG signals.

II. MECHANISM

A. Data processing

Part of the data of BCI Competition 2008 data set B is selected for analysis. Data collection of the data set is as follows:

At the beginning of each motor imagery test, the subjects were asked to sit at a position about one meter away from the display screen. The cross icon on the screen was changed into left or right arrow icon according to the different motor imagery task (left hand or right hand imagery). After the pattern lasted for about 1.25s, the experimenter performed motor imagery within 3-7s according to this prompt and obtained the corresponding motor imagery data. After this there will be a rest period of more than 1.5 seconds. Three bipolar recordings (C3, Cz, and C4) were recorded with a sampling frequency of 250 Hz. The recordings had a dynamic range of $\pm 100 \mu\text{V}$ for the screening and $\pm 50 \mu\text{V}$ for the feedback sessions [11]. They were band-pass filtered between 0.5 Hz and 100 Hz, and a notch filter at 50 Hz was enabled.

The original waveform usually have EOG and EMG influences [12]. During the EEG propagation process, the electrical signals needs to pass through several layers to reach the surface of the head to be captured, and therefore the location is vulnerable to interference. The captured signal may be affected by other movements of experiments, such as the swallowing movements. In order to reduce the noise of unknown interference, we reconstruct the EEG signal using the method of zero-setting reconstruction.

In addition to the possible errors generated by the subject, the waveform may also be affected by some artifacts—for example, the power line noise generated by the grid. The trap filters are used on the EEG data to remove noise such as power lines [13]. A trap filter can rapidly attenuate the input signal at a certain frequency point to achieve a filtering effect that prevents the signal at this frequency from passing. The waveform is processed to remove artifacts. If low frequencies are of interest, the curve can be obtained by low-pass filtering below the peak of the power line noise. To eliminate the slow drift, high-pass filtering can be used. A band-pass filter is used to complete the low-pass and high-pass filtering in one step.

B. ERD and ERS analysis

When a person does not exercise including imaginary movement or actual movement, the sensory or motor areas of the brain will show changes in the electrical activity of 8-30Hz [14]. The electrical activity of the brain during this idle period is reflected as μ waves in the motor cortex (9-13 Hz) and β waves in the visual cortex (18-30 Hz). When motion occurs, the idle rhythm amplitude will decay and can be detected near the wave and wave band [15]. A decrease in energy is called event-related desynchronization (ERD), and an increase in energy is called event-related synchronization. Time-related synchronization or time-related desynchronization potentials. The ERS and ERD are motor-related and are mainly located in the sensorimotor cortex [16]. ERD and ERS analyses of μ rhythm changes are as follows

$$ERD / ERS = \frac{A - R}{R} * 100\% \quad (1)$$

Here A is the EEG during motor imagery, and R is the energy value of the reference band before the motor imagery. The period of 1-2s before imagination is selected as R.

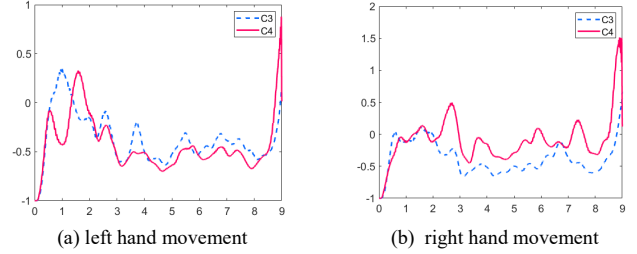


Figure 1. ERD and ERS analysis

In the figure 1, ERD occurs when people start to imagine motion (at about 2s), and ERS appears when the cue signal disappears (at about 8s). The figure shows that C3 and C4 reflect the μ rhythm changes of the contralateral side of the body respectively, such as ERD during exercise imagination and ERS energy changes after exercise.

Then, the paper analyses ERSP of event-related spectrum disturbance of each channel. It can be found that in the period of 3s-7s during the imaginary movement, the energy change is generally concentrated at about 10Hz, namely the μ rhythm. The average power obtained by C3 channel is larger than that obtained by C4 channel when imagining the left hand movement; the average power obtained by C3 channel is smaller than that obtained by C4 channel when imagining the right hand movement. This is also consistent with the ERD/ERS characteristics of the imagined motion EEG signal [17]: the unilateral limb motion or imagined motion produces event-related desynchronization potentials on the contralateral side of the brain and event-related synchronization potentials on the ipsilateral side of the brain.

C. FBCSP Model

FBCSP uses the relevant principles of information theory in machine learning to automatically select the characteristics of the relevant frequency bands obtained by the common spatial pattern (CSP) method through the method of mutual information [18].

It is divided into 4 stages: frequency domain filtering, spatial filtering, feature selection and classification. In the first stage, the measured value of the EEG signal is band-pass filtered into signals in multiple frequency bands; In the second stage, the corresponding CSP feature is extracted from each frequency band signal; In the third stage, the feature selection rule is used for automatic selection. The CSP features of the relevant frequency bands of the two tasks can be distinguished. In the fourth stage, the classification rule is used to classify the selected CSP features. The Common Spatial Pattern method is as follow.

We assume that the signal S is a matrix of $c \times (t \times n_\omega)$. Here c is the number of the channel, t is the number of samples

per channel, and n_ω is the number of class ω in the training data. If we do linear transformation for S

$$Z = W'S \quad (2)$$

$Z \in R^{c \times (t \times n_\omega)}$ is the signal after S has been spatially filtered.

$W \in R^{c \times c}$ is the mapping matrix of CSP, and W' represents the transposition character.

Feature selection based on mutual information is that, given an initial set F with d features, looking for a subset $S \subset F$ with k features so that the mutual information $I(S; \Omega)$ maximizes

$$I(X; Y) = H(X) - H(Y|X) \quad (3)$$

The entropy of the d-dimensional random variable $X = \{X_1, X_2, \dots, X_d\}$ is

$$H(Y|X) = - \sum_{x \in X} \sum_{y \in Y} p(x, y) \log_2 p(y|x) \quad (4)$$

p is a probability function.

In pattern recognition, the input features are usually discrete continuous variables, and the mutual information of the input feature X and category Ω is

$$I(X; \Omega) = H(\Omega) - H(\Omega|X) \quad (5)$$

Here $\omega \in \Omega = \{1, \dots, N_\omega\}$, and the conditional entropy is

$$H(\Omega|X) = - \sum_{\omega=1}^{N_\omega} p(\omega|X) \log_2 p(\omega|X) \quad (6)$$

Here N_ω is the number of class ω .

D. Adaboost algorithm

Adaboost algorithm is the abbreviation of Adaptive boosting, which is an iterative algorithm [19]. In every iteration, a new classifier is generated on the training set, and then the classifier is used to classify all samples to evaluate the importance of each sample. This article uses the Adaboost algorithm in a single-layer decision tree as a weak classifier. Specifically, the algorithm assigns a weight to each training sample. Each time a new classifier after training is used to label each sample. If a sample point has been classified correctly, its weight will be reduced, and the next data sampling will be carried out with this weight, and the probability of selection is reduced. If the sample point is not correctly classified, its weight is increased, and the next data sampling is carried out with this weight, and the probability of selection increases [20,21]. The higher the weight of the sample is, the greater the proportion of the next training will be. The entire iterative process does not stop until the error rate is small enough or iteration reaches a certain number of times. Adaboost algorithm process is showed in Figure 2.

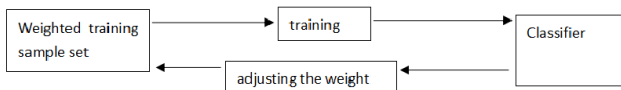


Figure 2. Adaboosting algorithm process

The adaptation of Adaboost lies in the fact that the samples misclassified by the previous basic classifier will be strengthened, and all the weighted samples will be used to train the next basic classifier again. At the same time, a new weak classifier is added in each round until it reaches a predetermined sufficiently small error rate or reaches the pre-specified maximum number of iterations.

E. Gradient Boosting algorithm

Gradient Boosting algorithm takes the residual as the training data of the next weak classifier, and each new weak classifier is built to reduce the residual of the previous weak classifier in the Gradient direction [22]. Gradient Boosting algorithm combines weak classifiers and replaces average voting mechanism with accumulation mechanism [23]. The processes of Gradient Boosting are as follows.

1) Initialization

$$f_0(x) = \arg \min_r \sum_{i=1}^N L(y_i, \gamma) \quad (7)$$

2) for $m=1$ to M

a) Calculate negative gradient

$$\tilde{y}_i = - \frac{\partial L(y_i, f_{m-1}(x_i))}{\partial f_{m-1}(x_i)}, \quad i = 1, 2, \dots, N \quad (8)$$

b) By minimizing the variance, use base learner $h_m(x)$ to fit $\tilde{y}_i$

$$w_m = \arg \min_w \sum_{i=1}^N [\tilde{y}_i - h_m(x_i; w)]^2 \quad (9)$$

c) Use line search to decide the step size ρ_m in order to make L the least

$$\alpha + \beta = \chi. \quad (10)$$

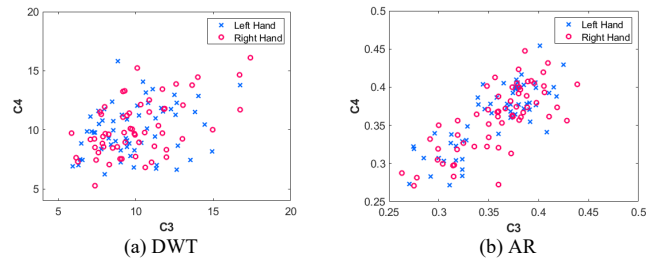
$$d) f_m(x) = f_{m-1}(x) + \rho_m h_m(x; w_m).$$

3) Output $f(m)$

III. RESULT

A. Feature extraction results

Five algorithms are used to extract features of EEG signals. To minimize individual differences among testers, we choose two sub-data sets from two person as training and testing data. The following is the two-dimensional feature point map of DWT, AR, PSD, CSP and FBCSP. The blue crosses represent the left-hand signals, and the red circles represent the right-hand signals.



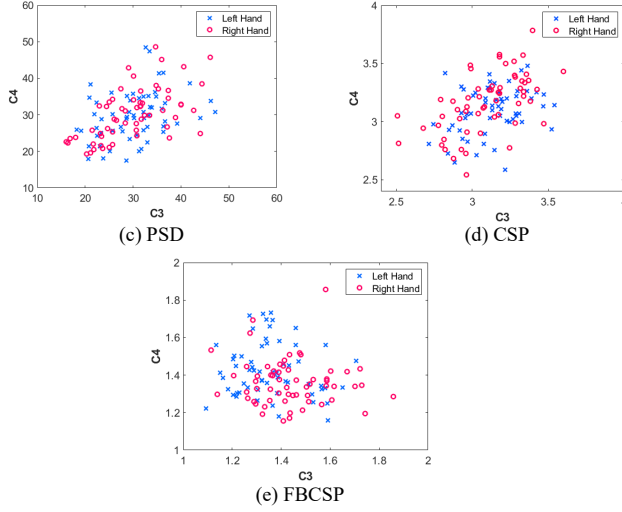


Figure 3. Feature extraction results

TABLE I. SVM CLASSIFICATION RESULT

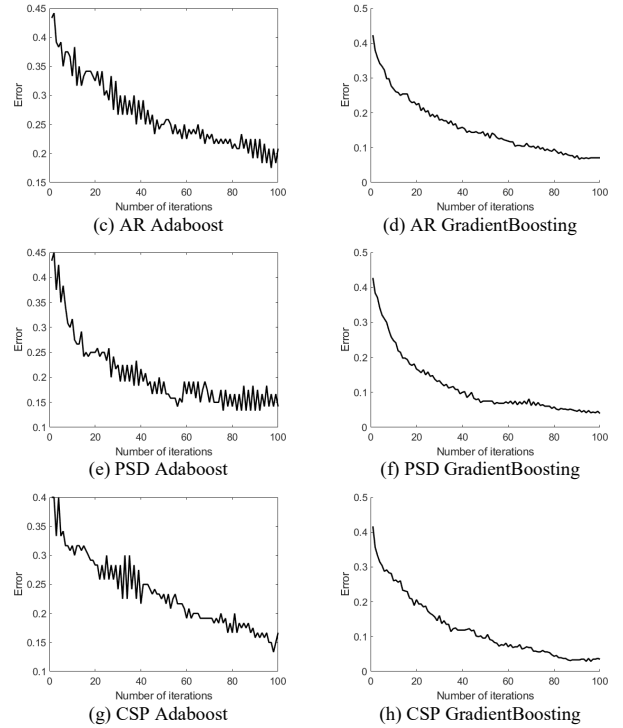
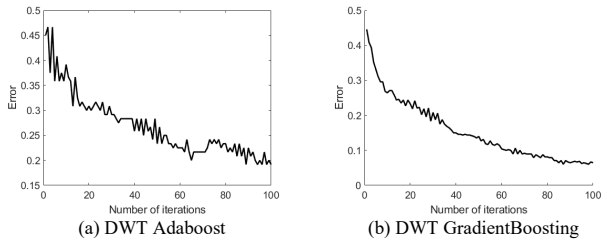
algorithm	classification result							
	iter	nu	obj	rho	nSV	nBSV	TotalnSV	accuracy
DWT	203	0.77	-147	0.011	110	66	110	45.71%
PSD	201	0.54	-78	0.031	117	17	117	50.00%
AR	60	1.00	-240	0.003	120	120	120	50.71%
CSP	68	0.88	-204	-1.389	108	104	108	52.14%
FBCSP	82	0.88	-202	-0.561	107	104	107	57.50%

Here iter is the number of iterations, nu is the parameter of the selected kernel function type, rho is the bias term of the decision function, obj is the minimum value obtained by the quadratic programming solution converted from the SVM file, nSV is the number of standard support vectors, nBSV is the number of support vectors on the boundary, and TotalnSV is the total number of support vectors.

When using the classification model SVM, the accuracy of four common algorithm DWT, PSD, AR and CSP is almost the same and unsatisfactory. However, the accuracy of FBCSP is much higher, and FBCSP method uses fewer support vectors, which means it has an advantage over those four algorithms.

Since the classification results of the SVM classifier of those algorithms are still not ideal, the ensemble algorithm Adaboost and algorithm Gradient Boosting are used.

C. Adaboost algorithm and Gradient Boosting algorithm



From the feature extraction results of figure 3, it can be seen that FBCSP method reduces the mixing of feature points of left and right hand signals, and feature vectors are more representative, resulting in better feature extraction effect.

B. The classification results

1) SVM algorithm.

The SVM essential model is a classifier that maximizes the interval in the feature space, and it is a two-class model [24]. The support vector machine is used to establish a classification model for the extracted two-dimensional features in figure 3, and the established model is used to calculate its corresponding accuracy on the test set [25]. The result of SVM classification with the five extraction algorithms is showed in Table 1.

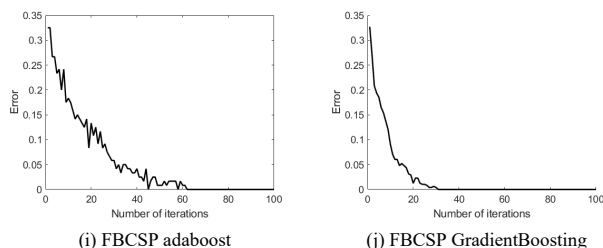


Figure 4. Adaboost and Gradient Boosting error rate with iteration

Figure 4 is about Adaboost and Gradient Boosting error rate with iteration. 100 iterations are performed to increase the weights of samples that are misclassified by the previous round of DWT, PSD, AR, CSP and FBCSP methods' SVM weak classifiers, and reduce the weights of the samples that are correctly classified to improve the classification accuracy. As for Adaboost, after about 100 iterations, all of five methods' errors decline to 10 to 20 percent, and they are much smaller than the previous SVM error rate which are about 50 percent. However, if we continue to let it iterate, the error might reach zero. In general, the more iterations are, the better the accuracy of classification will be. But at the same time, it means we need more time to iterate. Compared with Adaboost, the error rate line of Gradient Boosting algorithm drops more sharply with the increased number of iterations, and the accuracy of Gradient Boosting algorithm is higher. Besides, the error rate curve of Gradient Boosting is smoother, which means that accuracy fluctuates less. From figure 4, it can be found FBCSP has much lower error rate than other methods after 100 times of iterations.

IV. CONCLUSION

Filter Bank Common Spatial Pattern uses multi-band segmentation filtering, spatial projection and feature extraction to classify the EEG signals of left-right motor imagery, and its classification results are better than traditional DWT, PSD, AR and CSP algorithms.

AdaBoost uses exponential loss. The disadvantage of this loss function is that it is very sensitive to abnormal points, so it usually performs poorly on noisy data sets. Gradient Boosting has been improved in this aspect, so that any loss function can be used as long as the loss function is continuous and differentiable. After that, some more robust loss functions can be applied, making the model more resistant to noise. AdaBoost uses a strategy of increasing the weight of the wrong samples in the previous round of learning. In Gradient Boosting, the negative gradient is used as a measure of the error made by the previous round of the base learner. In the next round of learning, mistakes are corrected by fitting the negative gradient, so the error rate of Gradient Boosting usually is less than AdaBoost when having the same times of iteration.

Motor imagery by EEG signals is a branch of human motion posture recognition and motion prediction. This research topic has good application prospects as follows.

1) Intellect prosthesis tailored to the disabled community [26] enables prosthesis to move according to the user's consciousness.

2) The da Vinci robot [27] can have complex and precise surgery to be performed through minimally invasive methods.

3) Rehabilitation care [28] is used to calculate the movement trajectory of the exoskeleton through the analysis of the patient's movement posture and movement intention to assist the patient's movement. This can also improve the patient's participation.

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